

SHETH L.U.J AND SIR M.V COLLEGE
PRACTICAL :- 1
SUB :- Artificial Intelligence

CODE :-

```
import networkx as nx
import matplotlib.pyplot as plt

G = nx.Graph()

# Roads with distances
roads = [
    ("Malad West", "Link Road", 2),
    ("Malad West", "SV Road", 3),
    ("Malad West", "WE Highway", 4),

    ("Link Road", "Goregaon West", 2),
    ("SV Road", "Jogeshwari West", 3),
    ("WE Highway", "Malad East", 2),

    ("Goregaon West", "Oshiwara", 2),
    ("Jogeshwari West", "Andheri West", 3),
    ("Malad East", "Jogeshwari East", 3),

    ("Oshiwara", "Andheri West", 2),
    ("Jogeshwari East", "Chakala", 2),

    ("Andheri West", "Andheri East", 2),
    ("Andheri West", "Andheri Station", 1),

    ("Chakala", "Andheri East", 2),
    ("Andheri Station", "Andheri East", 1)
]

for u,v,d in roads:
    G.add_edge(u,v,weight=d)

# Position of each node
pos = {
    "Malad West": (0,5),

    "Link Road": (-3,3),
```

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```
"SV Road": (0, 3),  
"WE Highway": (3, 3),  
  
"Goregaon West": (-3, 1),  
"Jogeshwari West": (0, 1),  
"Malad East": (3, 1),  
  
"Oshiwara": (-3, -1),  
"Andheri West": (0, -1),  
"Jogeshwari East": (3, -1),  
  
"Andheri Station": (-1, -3),  
"Chakala": (3, -3),  
  
"Andheri East": (1, -5)  
}  
  
# Draw Figure  
plt.figure(figsize=(12, 8))  
  
nx.draw_networkx_nodes(  
    G, pos,  
    node_color="skyblue",  
    node_size=2500,  
    edgecolors="black"  
)  
  
nx.draw_networkx_labels(  
    G, pos,  
    font_size=8,  
    font_weight="bold"  
)  
  
# Draw all roads  
nx.draw_networkx_edges(  
    G, pos,  
    edge_color="gray",  
    width=2
```

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```
)

# BFS Shortest Path
bfs_path = [
    ("Malad West", "SV Road"),
    ("SV Road", "Jogeshwari West"),
    ("Jogeshwari West", "Andheri West"),
    ("Andheri West", "Andheri East")
]

nx.draw_networkx_edges(
    G, pos,
    edgelist=bfs_path,
    edge_color="orange",
    width=5
)

# Distance labels
labels = nx.get_edge_attributes(G, "weight")

nx.draw_networkx_edge_labels(
    G,
    pos,
    edge_labels=labels,
    font_size=8,
    font_color="red"
)

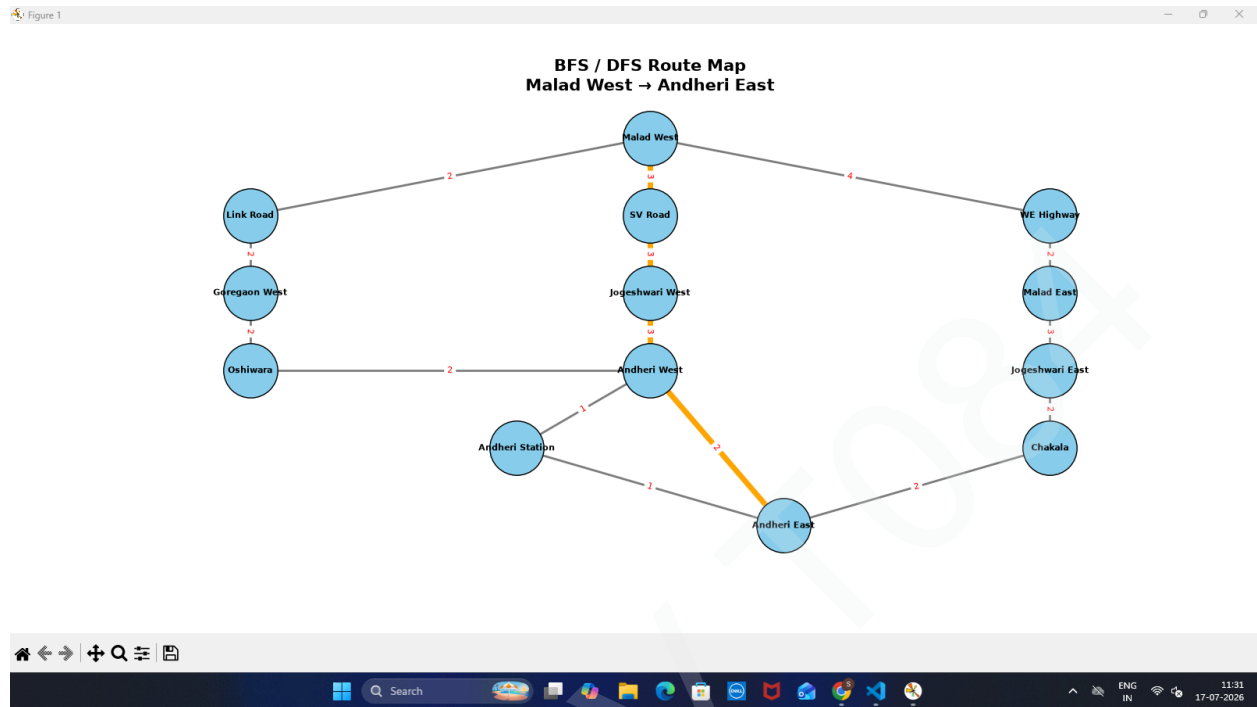
plt.title(
    "BFS / DFS Route Map\nMalad West → Andheri East",
    fontsize=15,
    fontweight="bold"
)

plt.axis("off")
plt.show()
```

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OUTPUT :-



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